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Submission Topic Assignment 4

Due date of submission 22/10/2020

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Group 05

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Assignment 3

Q1. (a) No.

Proof: Take $n_1 = 1$ and $n_2 = 5$.

Since $3n_1 + 1 \pmod{12} = 4 \bmod 12 = 4$ and $3n_2 + 1 \pmod{12} = 16 \pmod{12} = 4$

We have found different elements with the same image.

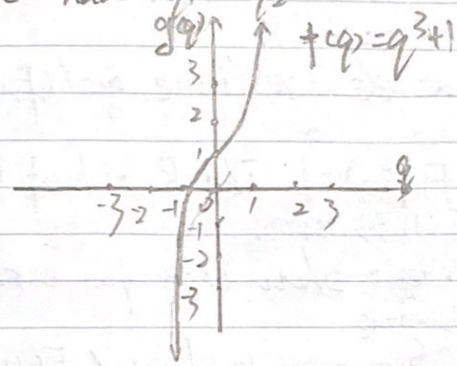
Thus f is not one-to-one. □

(b) Yes.

Proof: Suppose $q_1, q_2 \in \mathbb{Q}$ and $g(q_1) = g(q_2)$

Hence $q_1^3 + 1 = q_2^3 + 1$, so by subtracting both sides by 1

We have $q_1^3 = q_2^3$



Thus $q_1 = q_2$.

Therefore, $\forall q_1, q_2 \in \mathbb{Q}$, $g(q_1) = g(q_2) \Rightarrow q_1 = q_2$.

$\therefore g$ is one-to-one. □

(c) No.

Proof: Take $f(n) = 9 \in \mathbb{Z}_{12}$

Since $f(n) = 3n + 1 \pmod{12} = 9$

Hence $3n + 1 = 9 + 12k$ for some $k \in \mathbb{Z}$

$3n = 8 + 12k$ for some $k \in \mathbb{Z}$.

$n = \frac{8}{3} + 4k$ for some $k \in \mathbb{Z}$.

Since $n \in \mathbb{Z}_{12}$, there does not exist an element in the domain with image 9.

Therefore f is not onto. □

(d) No.

Proof: Take $g(q) = 11 \in \mathbb{Q}$

So $g(q) = q^3 + 1 = 11$

$q^3 = 10$

$q = \sqrt[3]{10} \in \overline{\mathbb{Q}}$

Since $q \in \mathbb{Q}$, there does not exist an element in the domain with image 11. $\square$

Q2. (a) Proof: Suppose $\gcd(a, b) = d$

Since $a, b \in \mathbb{Z}$, $a = dk$ for some $k \in \mathbb{Z}$, $b = dl$ for some $l \in \mathbb{Z}$

$$a + b = dk + dl$$
$$= d(k+l)$$

Thus $\gcd(a, a+b) = \gcd(dk, d(k+l))$
$$= d$$

Therefore $\gcd(a, b) = \gcd(dk, a+b) = d$. $\square$

(b) Proof: Let $F(n)$ be the predicate $\gcd(F_n, F_{n-1}) = 1$

Basic Step: $F_1 = F_0 = 0 = 1 + 0 = 1$

The LHS of $F(1)$ is $\gcd(F_1, F_0) = 1$. The R.H.S of $F(1)$ is 1.

Thus $F(1)$ is true.

Inductive Step: We must show that for every $n \geq 1$, if $F(n)$ is true then $F(n+1)$ is true.

Inductive Hypothesis: Suppose $n \geq 1$ and $F(n)$ is true.

This means $\gcd(F_n, F_{n-1}) = 1$

Now consider $F(n+1)$.

From (a), we get for any integers a and b, $\gcd(a, b) = \gcd(a, a+b)$

Therefore, LHS of $F(n+1) = \gcd(F_{n+1}, F_n)$
$$= \gcd(F_n, F_{n-1})$$
$$= \gcd(F_n, F_n)$$
$$= 1 \qquad \square$$

Q3 Proof: ρ is reflexive.

Consider $a, b, x, y \in \mathbb{Z}$, $b, y \neq 0$.

Then $ay = bx$

$$ay - bx = 0.$$

And so $(a, b) \rho (x, y)$.

ρ is symmetric.

Suppose $(a, b) \rho (x, y)$

Then $ay = bx$ and so $xb = ya$

Hence $(x, y) \rho (a, b)$

ρ is transitive.

Suppose $(a,b)\rho(x,y)$ and $(x,y)\rho(k,l)$.

Then $ay=bx$, $xl=yk$,

and so $ayxl=bxyk$

Hence $al=bk$, and so $(a,b)\rho(k,l)$.

So ρ is an equivalence relation. $\square$.

Q4. (a) Yes

Proof: ① $f(A\cup B)\subseteq f(A)\cup f(B)$

Let $y\in f(A\cup B)$, then $\exists x\in A\cup B$ such that $f(x)=y$.

Thus $\exists x\in A$ or B such that $f(x)=y$

Since $f(x)=y$, $y\in f(A)$ or $f(B)$

Thus $y\in f(A)\cup f(B)$

Therefore, $f(A\cup B)\subseteq f(A)\cup f(B)$.

② $f(A)\cup f(B)\subseteq f(A\cup B)$

Let $y\in f(A)\cup f(B)$, then $\exists x\in A$ or B such that $f(x)=y$

Thus $\exists x\in A\cup B$ such that $f(x)=y$

Since $f(x)=y$, $y\in f(A\cup B)$

Thus, $f(A)\cup f(B)\subseteq f(A\cup B)$.

So $f(A\cup B)=f(A)\cup f(B)$. $\square$

(b) No.

Proof: ① $f(A\cap B)\subseteq f(A)\cap f(B)$

Let $y\in f(A\cap B)$, then $\exists x\in A\cap B$ such that $f(x)=y$

Thus $\exists x\in A$ and B such that $f(x)=y$

Since $f(x)=y$, $y\in f(A)\cap f(B)$

Thus, $f(A\cap B)\subseteq f(A)\cap f(B)$.

② $f(A)\cap f(B)\subseteq f(A\cap B)$

Define $A=\{-1,-2\}$, $B=\{1,2\}$, $f(x)=x^2$

Thus $f(A)=\{1,4\}$, $f(B)=\{1,4\}$

Therefore, $f(A)\cap f(B)=\{1,4\}$

$A\cap B=\emptyset$

Thus $f(A\cap B)=\emptyset$

Therefore $f(A)\cap f(B)\not\subseteq f(A\cap B)$

So $f(A\cap B)=f(A)\cap f(B)$ is not always true. $\square$